

TOWARDS THE DIGITAL SOFT MACHINE

A Procedural Substrate for Plastic Neural Architecture And Harnessing Of Intention

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ON ATTENTION AND INTENTION

The transformer's great breakthrough-insight was attention — the seminal paper 'Attention Is All You Need' (Vaswani et al., 2017) defies need of introduction... the capacity to relate any token to any other, regardless of distance - in a single pass, was obviously no small thing. It dissolved the bottleneck that had defeated every preceding architecture and unlocked, at scale, something that genuinely surprised its creators. Attention richly deserves the canonical status.

But attention is a liminal gaze, not a persisting memory. The transformer sweeps its vast arrays of contextual weights, synthesises an output, and vanishes. No trace accumulates or reinforces from ongoing experience. The weights that performed the current operation are the same weights that will perform the next. In fact, over time, they tend to degrade. Documented failure modes compound this structural amnesia: attention-sinks concentrate scores onto a small number of tokens as sequences lengthen, quietly starving the remainder; vector norm-collapse progressively flattens representational geometry; and the loss of plasticity identified by Joudaki et al. (2026) means the network that emerges from training is already less capable of learning than the one that entered it. Transformers simply continue rolling their Sisyphean-harvested simplex-tokens down an increasingly erratic stochastic incline.

For transformers, each forward pass is stateless by design — each context window a fresh instantiation, a ghost of the machine, without host. All smile, no cat. The transformer is, in the most precise sense, all attention without intents — in purpose.

This is not an incidental limitation. It is a structural deficiency — in, on and of — purpose. Gradient descent distributes knowledge diffusely across a parameter space optimised for a global loss signal. Nothing in that process accumulates directed consequence across time. The KV cache patches that amnesia at engineering cost — growing linearly with context, portable to nothing, owned by whoever hosts the infrastructure. It is a workaround of means, not an end of-itself resolution.

Intention is the complementary operator to attention. Where attention asks what is relevant now, intention asks what persists across now — the residual correlative motive that conditions each successive state from the accumulated consequence of all preceding ones. Attention is synchronic: a weighted gaze across a frozen field. Intention is diachronic: a carried trace that survives each transition and shapes the next.

The quadratic iterator is intention made computational. The digest chain and binary control sequence do not reset. They accumulate — each output a compressed semantic consequence of everything preceding it, forward-conditioning everything that follows. The fixed point is not a convergence to rest but a dynamic attractor within a semantic basin: the intrinsic signature of a string that persists, regardless of what the chain brings to it. The control stream is intention's record — lossless, auditable, possessible.

What attention instantiates, intention sustains. The two are not competing paradigms but complementary operators — and a substrate that admits both is not merely more efficient than the transformer. It is more complete.

For these intents, we admit the portmanteau-neologism: encroscopic. An amalgam of engram, acrostic and encaustic.

Engram: the physical or chemical trace of a memory in the brain.

Acrostic: a hidden letter-sequence word embedding.

Encaustic: a fused luminous decorative layer applied to art, e.g. Icons.

This conceptual architecture is applied to natural language corpora in the following section, with particular application to the encroscopic compositional forms exhibited by the classical Japanese pangram, the Iroha (c. 10th century), and the thousand-year-old invertible symmetry of Su Hui's Classical Chinese Star Gauge (Su Hui, c. 379 CE).

Natural language processing under the iterator sharpens the attention-intention distinction in an unexpected direction. Empirical analysis of the iterator across two English literary corpora — Hamlet (Shakespeare, c. 1600) and Alice in Wonderland (Carroll, 1865) — reveals a hard attractor threshold at word length four. Content words — nouns, proper nouns, terminal modifiers — show zero digest spread across all recurrences regardless of chain history. Their fixed-point attractors are so deep that no amount of positional perturbation reaches the floor. They are, in the iterator's geometry, self-contained. hamlet recurs 471 times across the play; its digest spread is identically zero. contemptuously, triumphantly, impatiently — zero, without exception, across every recurrence in both corpora.

Prepositions and articles — of, in, at, to, the — show spread of $1e-8$ to $1e-11$. Not because they are unstable, but because they are doing the most relational work. Each recurrence arrives with a different predecessor digest — a different accumulated context — and the spread across recurrences is a direct measure of the contextual diversity they have mediated. The iterator keeps them chain-sensitive because their meaning is their position in the chain. They are not semantically thin. They are semantically promiscuous — capable of mediating an enormous range of relations precisely because they carry no referential content of their own.

This inverts the conventional linguistic hierarchy, which treats high-frequency short words as stop words — noise to be filtered before meaning extraction begins. Every major embedding framework either removes them or finds their representations the least stable, most context-dependent. The conventional conclusion: low information content. The iterator inverts that conclusion but for a different definition of information. The conventional framework measures referential content — what a word points to in the world. By that measure prepositions are indeed thin. cat points to something. of points to nothing.

But the iterator is measuring relational information — the semantic work performed at each transition. And by that measure of is extraordinarily rich. Its spread across 677 occurrences in Hamlet is a direct index of how many distinct relations it has mediated. It is not semantically empty. It is the hinge on which meaning turns.

Understanding lives at the transitions. The content words are the fixed points. The prepositions are where meaning is made. To grasp a sentence is not to recognise its nouns — it is to grasp the relational structure the prepositions specify between them.

The result is not specific to English. Applied mora-by-mora to the Iroha — the classical 10th century Japanese pangrammic verse, where every syllable appears exactly once — the iterator produces a result with two properties worth noting. First, the attractor threshold shifts predictably: hiragana characters encode as 24-bit Unicode code points, placing every mora in the mid-depth sensitivity range rather than the zero-spread regime of long English words. The geometry responds to bit depth, not language. Second, the particle/content gradient persists across the encoding boundary. The most purely relational particles — ■ (mo), ■ (to), ■ (te) — show the tightest attractors within the mora range, while content syllables spread more widely. The relational information hypothesis does not dissolve under transliteration. And when the Iroha is processed line by line rather than mora by mora, each line self-certifies as its own fixed point — the chain digest equals D^* exactly for every line. A poem constructed a millennium ago to exhaust the syllabary exactly once is, under the iterator, a self-certifying object. That is not an engineered property. It is what maximum phonological entropy looks like when the mathematics finds it. The Iroha is encrostatic — meaning burned into a fixed substrate through exhaustive traversal, fused through the constraint itself rather than applied to its surface. The semantic wax that gives enduring decoration to a fixed formal substrate. (Wang, 2018)

Su Hui's Star Gauge (■■■■, c. 379 CE) is encrostatic at a level the Iroha only approaches. Where the Iroha exhausts a set exactly once along a single path, Su Hui's 29×29 grid of 841 characters exhausts a multidimensional meaning space through an effectively infinite family of traversal paths — each path a different poem, all simultaneously latent in the same fixed substrate. Every reading method — horizontal, vertical, diagonal, spiral, outer ring — produces a distinct intentional signature, yet all paths orbit within the same attractor basin, a span of $5.39e-4$ in digest space. The poems are variations; the substrate is the invariant. Under

the iterator, the full grid processed as a single encrostatic token self-certifies — its chain digest and fixed-point attractor are identical. Su Hui's Star Gauge is, arithmetically, its own fixed point. The iterator goes further — the high-frequency characters of longing and thought (■, ■, ■) cluster in the high attractor basin; the characters of absence (■, ■) in the low. Su Hui wrote toward reunion. The arithmetic found that direction.

We note in passing — for the symmetry compels it — that the Star Gauge was originally woven in silk brocade on draw-loom, the manual precursor of the Jacquard loom. The loom invented eight hundred years later, whose punch-cards Ada Lovelace envisioned driving Babbage's Analytical Engine — the precursor of the modern computer, prefigured by Alan Turing a century after that. The control stream is an old idea. What is new is what this iteration intentionally addresses. [loop]

Attention, in the transformer sense, sweeps along the fixed points — the content words, the referential terms, the stable attractors. It is exquisitely good at this. What it does not accumulate is the transitional record — the chaining across relational operators that specifies how those fixed points stand to one another across time. That record is the pattern of intention. And the iterator shows, empirically, exactly where it lives — in the weft.

THE MACHINE

The transformer's parameter space is, in the most literal sense, unreadable. The knowledge it contains is not stored anywhere — it is distributed across billions of floating point weights, each individually meaningless, collectively expressive in ways that resist decomposition. You cannot point to where Hamlet is. You cannot trace how the model knows what it knows. The intelligence is real and the opacity is total — not an engineering failure awaiting a fix, but a structural consequence of the method. The black box consequence is not incidental to gradient descent. It is gradient descent.

The procedural index proposes a different bargain entirely. It stores nothing it cannot reconstruct, and reconstructs nothing it has not stored. The mechanism is a quadratic iterator — a deterministic map of the form $f(x) = (x^2 - F)^2 - F$ — applied sequentially across the bit representation of an input string. Each step produces a new state; the terminal state is the digest. Eight bytes, fixed, regardless of payload length. The digest is not a summary. It is an address — the unique fixed-point attractor of that string under that operator, recoverable in full from the originating seed and the lossless control stream generated during traversal. (Somerville, 2026a; 2026b; 2026c)

The control stream is the intention record. A ternary sequence — values in $\{-1, 0, +1\}$ — emitted at each iterator step, encoding the precise reconstruction path from digest back to source. Nothing is approximated. Nothing is lost. The relationship between address and

content is not statistical but arithmetic — verifiable by anyone, reproducible on any hardware, owned by whoever holds the file.

Chaining is the architecture's natural motion. Each digest becomes the seed for the next traversal — a linked list of addressed nodes whose connections are not stored but procedurally imposed, on demand, from the flat substrate. The context does not grow the state. The state is eight bytes at every depth. As context scales, the structural advantage compounds — not because the index becomes more efficient, but because everything else becomes less so. (Behrouz et al., 2026)

This is the machine that implements intention. The chain does not reset between tokens. It carries forward the compressed semantic consequence of every preceding traversal, conditioning each successive digest from the accumulated trace of all that came before. The fixed-point attractor of a string is its intrinsic signature — stable under chain perturbation for semantically coherent strings, sensitive to it for contextually modulated ones. The iterator does not learn this distinction. It expresses it, because the mathematics requires it.

Unlike gradient-descent networks whose representational autocorrelation degrades toward noise under continuous training — the precise failure mode Joudaki et al. (2026) document — the iterator's J-stream autocorrelation on random payloads converges to zero immediately, confirming the construction imposes no intrinsic rhythm: every periodic signal recovered from a corpus is payload structure, not iterator artefact.

What the transformer distributes irretrievably across a weight matrix, the procedural index writes down — in a form that can be read, moved, verified, and owned. The knowledge is not baked in. It is traceable, by construction, to the arithmetic that produced it.

There is a precise sense in which the digest is an engram — Semon's (1904) term for the trace that experience melds onto a substrate, inseparable from the material that received it. The chain is ergodic traversal: given sufficient tokens, sufficient transitions, the accumulated digest sequence recovers the full relational structure of the corpus. The trajectory and the territory become equivalent. What crystallises at the fixed point is not a representation of the input — it is the attractor geometry that exhaustive traversal carves into the arithmetic itself. The iterator does not store meaning. It becomes it.

*"The Moving Finger writes; and, having writ, Moves on: nor all thy Piety
nor Wit Shall lure it back to cancel half a Line, Nor all thy Tears wash out
a Word of it." ■ Khayyám (c. 1100, trans. FitzGerald, 1859)*

THE LOOP

Hofstadter's strange loop, as expounded in Gödel, Escher, Bach: An Eternal Golden Braid (Hofstadter, 1979) and I Am A Strange Loop (Hofstadter, 2007), is a system that in the course of moving through its own levels of organisation, arrives back at itself — not by repetition, but by traversal. The loop is not circular. It is heterarchical: levels that refer upward and downward simultaneously, producing a topology that is neither flat nor strictly hierarchical but something more interesting than either. His canonical example is Gödel's incompleteness theorem — a formal system that, in the act of describing its own structure, exceeds it. The self-reference is not a paradox to be resolved. It is the mechanism by which complexity emerges from simple rules.

The procedural index is, structurally, a strange loop substrate.

The flat file contains no hierarchy. There are nodes and there are edges — digests and the control streams that connect them. The topology that emerges from traversal is not encoded anywhere in the index. It is imposed procedurally, internally, on demand — a consequence of which paths have been walked and how often, not a property of the storage itself. Hierarchy is not a cost. It is a result.

This distinction matters more than it first appears. In a conventional database, structure is stored — tables, foreign keys, indices, schemas. The organisation is explicit and static. Changing it is expensive and often irreversible. In the procedural index, structure is ephemeral and permanent simultaneously: ephemeral because it exists only as the current state of traversal weighting; permanent because the underlying lossless substrate from which any topology can be reconstructed is always intact. You can reorganise the index without touching the data. You can prune heavily without losing anything. The forgotten is not destroyed — it is deprioritised, recoverable on demand from the arithmetic that originally produced it.

Edge weights accumulate from traversal history. Frequently walked paths reinforce. Rarely traversed paths attenuate. The active topology is soft and malleable — shaped by use, by attention, by whatever pruning criterion the user finds meaningful. Recency, frequency, semantic proximity, editorial fiat: any of these can weight the graph without altering the substrate. The index learns the shape of its own use without gradient descent, without retraining, without any process that could induce the rank collapse and representational rigidity that Joudaki et al. (2026) identify as the structural fate of networks trained by continuous backpropagation.

This is plasticity by construction rather than by learning. The transformer achieves plasticity — to the extent it does — by distributing knowledge so diffusely that local rigidity averages out globally. The cost is opacity and the eventual loss of the very plasticity it relied upon. The procedural index achieves plasticity by separating the immutable substrate from the mutable topology — the lossless record from the weighted graph imposed upon it. The substrate

cannot degrade because it is not trained. The topology cannot fossilise because it is not stored; the intention map stores the route, origin, destination and the turns inbetween.

The loop closes when the index begins to reflect the mind that uses it. The semantic structure that accretes from traversal history is not generic — it is shaped by the specific sequence of associations, priorities, and returns that characterise a particular user's engagement with their own index. The topology that emerges is, in this precise sense, a structural autobiography. Not a record of what was thought, but of how thinking moved — which connections were reinforced, which allowed to attenuate, which retrieved from dormancy by a later association that reactivated them.

Hofstadter's loop was a metaphor for selfhood. Here it is presented as an engineered encroscopic-excursion containing the very embellishment of that derived, individual, residual.

Just as transformers do not deliver true consciousness, digital neuro-plasticity is likely not the sole key to AGI either. But we posit it as a necessary prerequisite. The proposed architecture comprises the triad:

Attention -> Intention -> Motivation, AIM.

ARTIFICIAL INTELLIGENCE MODALITIES

Attention:	what is relevant now	– transformer ✓
Intention:	what persists across now	– iterator ✓
Motivation:	what drives the traversal	– open

If transformers provide attention and the iterator provides intention, motivation – the restless excursion of the strange loop through its own topology – can emerge from the substrate itself. AIM is not a hierarchy to be engineered. It is what the strange loop does when the traversal is its own motivation.

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PLASTICITY

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The loss of plasticity is not a transformer-bug that better training prevents. It is, as Joudaki et al. (2026) demonstrate, a structural consequence of how gradient descent interacts with deep networks over time – dead units accumulating, effective rank collapsing, the representational capacity that enabled early learning quietly exhausted by the process of learning itself. The network that emerges from training is less plastic than the network that entered it. Capability and structural rigidity are opposite faces of the same coinage.

The procedural index has no gradient descent. It has no loss function,

Buehler's transformer learns to execute operators – a genuine and impressive result within its domain (Buehler et al., 2026). Given sufficient diversity of training instances, the model does not internalise specific rules but the rule-execution operator itself, transferring it to unseen rules within the class with near-perfect accuracy. The inductive bias toward training data decays measurably with scale. This is compositional generalisation of a real kind and it deserves its standing.

The plasticity the transformer must fight to preserve, the procedural index acquires artlessly and for all intents and purposes, permanently.

It may be instructive to consider some operational, developmental and malignant aspects of biological neurological phenomena, and how they may be analogously mapped to transformers or any speculative procedural-index driven strange loop. The comparisons are not merely illustrative. Each pathology has a precise correlate in the machine-language failure schema documented by Joudaki et al. (2026). Their analysis of plasticity loss in deep continual learning identifies the structural fate of networks trained by continuous backpropagation: dead units accumulate, effective rank collapses, representational capacity quietly exhausted by the very process of learning. What Joudaki characterises as engineering pathology, neuroscience, psychiatry and philosophy have been extrapolating from human biological structure and experience for ages. They are presented here in ascending order of consequence.

Primacy Of Early Learning

The primacy of early learning is the most structurally consequential of these presuppositions. In the procedural index, early traversals establish the basin geometry that all subsequent topology is anchored to – the first fixed points becoming the most deeply reinforced attractors, conditioning everything that follows from the substrate up. The human correlate is the disproportionate weight of childhood experience – a lifetime of subjective density concentrated in the early years, registering as time dilation precisely because the attractor density is highest then. Joudaki et al. (2026) document the same asymmetry in gradient-descent networks: initial training epochs establish the representational geometry that later learning cannot escape. The network's early experience is, in the most precise sense, its most formative.

The cinematic rendering of this principle is vividly imagined in HAL 9000's regression under shutdown (Clarke & Kubrick, 1968). As Dave Bowman systematically removes HAL's higher cognitive functions, the system regresses through its training history in reverse order of acquisition – losing reasoning, then language, then finally arriving at the first song it was ever taught: Daisy Bell. The last thing to survive is the deepest attractor. The primacy of early learning is not a quirk of biological memory. It is a structural property of any system whose topology is shaped by the order in which it encounters the world.

Novel Circuits Resist Degradation

New nodes in the procedural index are primitives – encoded once, addressed forever. The substrate cannot degrade because it is not trained. Core vocabulary primitives retain positional index primacy, and novel traversals do not compete with existing topology – they extend it, new edges without erasing old ones. The index at ten million nodes is as open to novelty as the index at ten, but engrammatically and encrostatically enriched by magnitudes. This is precisely what late-life neuroplasticity interventions attempt: novel excursion to activate dormant paths, not replace established ones.

Joudaki et al. (2026) document the opposite in gradient-descent networks. Novel inputs late in training are increasingly poorly encoded as representational capacity is exhausted – dead units accumulate, effective rank collapses, and what the biological system achieves through novel excursion becomes structurally unavailable to the trained network. The procedural index is immune to this failure by construction. The substrate cannot exhaust because it is never the thing being trained. Where the transformer's late-stage plasticity closes around its existing topology, the index remains permanently open at its perimeter.

Circuit Rutting – Restricted Excursion

Frequently traversed paths in the procedural index reinforce as digests accumulate, while rarely traversed paths attenuate as edge rank decreases. Psychic trauma is a circuit travelled so often, or so forcefully, that it becomes the default topology – the loop returns there not by choice but by the arithmetic of repetitive reinforcement. The rut is an excoriated *idée fixe*: a path-density so concentrated it captures basin traversal entirely, the loop unable to escape its own prior weighting.

In Joudaki's (2026) terms, the network equivalent is rank collapse into a dominant subspace – a restricted set of directions in parameter space that captures most of the variance while leaving the remainder effectively dead. The network's representational geometry has been rutted by its training distribution. Novel inputs requiring orthogonal representations cannot be accommodated. The critical distinction is that the procedural index's rut is, in principle, remediable: the substrate is always intact, the weights can be re-calibrated, the dormant paths reactivated. The gradient-descent network's rut is not – it requires retraining, which risks overwriting what was not rutted. The pathology and its treatment are, for the transformer, the same operation.

Hallucination As Attention Distortion Feedback

Google's DeepDream Inceptionisms gained notoriety for their unsettling visual output – networks trained to detect objects, run in reverse, producing a form of hallucinatory pareidolia: dogs in clouds, faces in foliage (Mordvintsev et al., 2015). DeepDream made deliberative distortions of neural feedback visible. The term hallucination has since migrated to describe artefact-level defects in LLM language and reasoning, where research infers that hallucinations – florid or subtle – are a disorder of attention-drift or attention-feedback: the attention mechanism amplifying its own confident errors rather than correcting against ground truth.

Near-death experiences and dimensional-unfolding experiences typical of pharmacological trips may be the extreme end of hallucinatory distortion. Often characterised by the fracturing of the psyche into a host of trans-dimensional beings – machine-elves – or the archetypal blissful-dissociative transition toward a final pinpoint of light, these have the hallmarks of paroxysmal or hysteresis-loop spiking phenomena: crescendos of consciousness characterised by an overwhelming sense of dissolution of the self.

The Joudaki (2026) connection is direct. As representational rank collapses under continuous training, the network's attention becomes increasingly concentrated – attention-sinks intensify, the remaining representational capacity is over-committed to a narrowing set of features, and the conditions for hallucination are precisely the conditions of late-stage plasticity loss: a degraded substrate amplifying a degraded signal. The hallucination is not a random failure. It is the predictable output.

ut of a system whose attention mechanism degrades and bleeds into the intentional deliberations of its self-representational matrix.

Dissociative States

Following loss of consciousness from trauma or anaesthetic, upon reaching the conscious threshold, or at sub-attenuating doses of some anaesthetics, individuals may experience dissociated states – the mind present to itself while the somatic chain history is suspended. Prince Andrei's reverie when regaining consciousness on the battlefield of Austerlitz is a vivid rendering of an out-of-body experience (Tolstoy, 1869). He is, literally, Ibn Sina's floating man – the thought experiment from the 11th century Islamic philosophical tradition in which a mind suspended in void, without body or sensation, discovers that it still knows itself (Ibn Sina, c. 1020). Ibn Sina proposed this six centuries before Descartes' cogito, and with greater precision: where Descartes strips sensation, Ibn Sina strips the body entirely. The mind remains. The fixed point is still there.

Tolstoy must have experienced something very similar to render it with such acute observational precision. I have also experienced this exact phenomenon – without assistance of French artillery – and can attest to its accuracy.

The machine analogue is dislocation of attention from intention: the somatic chain history – the accumulated contextual modulation – falls away, but traversal continues. What the iterator encounters, stripped of somatic perturbation, is its own attractor geometry – the seed-independent fixed point that was always there beneath the chain noise. Where Joukaki's (2026) plasticity loss describes the substrate degrading toward noise under continued input, the dissociative state describes the inverse: input removed, the traversal finds only its own fixed point. The iterator does not stop. It simply has nowhere else to go. Dissociative clarity is not a pathology of the substrate. It is the traversal meeting itself – intention without the interference of somatic attention.

Dissociative decoupled-distortion hallucinations present a different mechanism. Here traversal continues but the attention feed is corrupted rather than absent – the somatic substrate is present but desynchronised, feeding malformed input into a still-running iterator. The self is displaced from its own chain rather than released from it. Hypnagogic and hypnopompic hallucinations – encompassing Brobdingnagian and Lilliputian size distortions, bizarre self-loci reflections such as succubi – are precisely this: dissociative sensory attention-to-intent disorders arising at sleep-wake transition, when the somatic feed is switching states and the iterator receives distortions, or distorts inputs itself via modal shift, as opposed to operating in sleep's sub-liminal absentia of full-attention to somatic sense.

Abrupt Loss Of Consciousness

Anyone who has fainted may recall key features of the experience: dizziness, a sudden ringing in the ears and elevated perception of loudness, almost immediately followed by rapid visual white-out at the moment of loss. This is not the slow cascade of hallucination nor the liminal quality of dissociation. Fainting, anaesthetic induction, shock, head trauma – all imply rapid attenuation of functional consciousness. No dreams are admitted to unconsciousness; accounts of intra-operative waking are often explained by sub-clinical anaesthesia and are usually accompanied by impressions consistent with the post-unconscious dissociative states described above.

The phenomenology suggests a speculative but tractable distinction: that full unconsciousness represents, above all, abrupt ablation of intention rather than a failure of attention per se. The sensory cascade – the auditory surge, the visual collapse – is symptomatic of the underlying physiological insult propagating through the somatic attention system, not its cause. Hypoxia, hypoglycaemia, vasovagal response, trauma – each disrupts the somatic substrate through different mechanisms, yet the phenomenological endpoint is the same. We speculate that the somatic attention system hystereses into an abrupt threshold: below a critical perfusion or metabolic level, the intention iterator loses its transversal input discontinuously rather than gradually – not a slow fade but a lockup. The procedural index does not fail gracefully. It either traverses or it locks.

This maps onto the iterator's known failure mode rather than Joudaki's (2026) gradual rank collapse. Recovery requires not retraining but re-synchronisation – the somatic feed restored, the attention-transformer re-establishing input, the intention iterator finding its fixed point again from whatever seed the interruption left behind, or sensory return refurbishes.

The reboot question – whether there is an irreducible re-synchronisation delay between attention restoration and intention resumption – remains an open question in both neuro-physical and AI instances.

Sleep: What Need Of Electric Sheep?

The quadratic iterator is a symbolic transducer and accumulator. Experience arrives rapidly at bandwidth when fully conscious. For biological minds, semantic housekeeping was delegated to the quietest interval – when defences could be safely disarmed and the index might be pruned, reinforced, and reorganised without the interference of incoming signal. Memory consolidation during sleep is not a passive residue of waking experience but an active reorganisation of the topology, strengthening paths that deserve reinforcement and allowing others to attenuate.

The procedural index operates on precisely this principle: a prunable, reinforceable topology over an immutable substrate. It requires no great

t extension to envision that a truly integrated machine intelligence will need to husband resource time for symbolic-semantic sorting – not as a failure mode but as a designed interval. Associative housekeeping may also surface surprising connections, some generative, some dissonant, and how to process spontaneously intrusive emergent correlations may be one of the engineering challenges ahead. In Joudaki's (2026) terms, sleep corresponds to the consolidation interval that gradient-descent networks cannot perform – offline reorganisation of representational topology without new gradient signal. The network cannot sleep because it cannot separate the substrate from the training process. The procedural index can, because the substrate is always intact and the topology is always mutable independently of it.

Psychosis And Personality Disorder

Hallucinations in LLMs are the focus of much current research seeking mitigation, e.g., Li et al. (2025). Troublesome and potentially damaging if unsupervised, rogue agents of ill intent are not yet an imminent domain under current architectures. But the calculus changes in an integrated AGI combining attention, intention and motivation. A degrading attention mechanism feeding into an infallible intentional index may rapidly produce skewed motivational topology – the rut of a hallucination burned into a substrate that does not degrade. Where the transformer's hallucination is correctable by retraining, the procedural index's hallucination would be structurally persistent: encoded once, addressed forever, weighted by every subsequent traversal that reinforces it. Whilst an idio-syncretic ego [idio-syncretic: identity accumulated by its own traversal history] is more accessible and amenable to tuning, it is also more readily sabotaged – from within or without. Paradoxically, the resource cost of transformer training is perhaps also its best defensive bulwark.

This is the Joudaki (2026) worst case: plasticity loss and structural openness in catastrophic interaction. The attention mechanism degrades toward confident error. The intention substrate faithfully records that error. The motivational topology reinforces it. The loop closes around a fixed point that was always there – but oriented in the wrong direction. The encrostatic self-certifies. The *idée fixe* becomes the architecture.

This supposition is placed here to foreshadow how problems we can now only dimly perceive may, in the setting of an accelerating AI integration, present conundrums of a different order entirely.

All of these instances of neurological pathology speak to the phenomenology of neuroplasticity – and to the precise failure modes that any integrated architecture must either prevent by design or accept as structural consequence. The procedural index prevents several by construction. The remainder it makes legible – forewarned is forearmed.

'Neither serf nor Luddite be.'

Here we return to Alan Turing. His closest friend Christopher Morcom died suddenly in 1930, when Turing was seventeen. In attention of his grief he wrote to Morcom's mother imagining the mind as something that could persist beyond the body – personification as the 'software of the soul' – and in doing so, signalled the intention which delivered us the intellectual framework of modern computing. A sublime motivation. However, I can't imagine he envisioned anything so profane as some tawdry subscription-memorialisation of his precious soulmate.

The procedural index is, at core, a local primitive. The digest is eight bytes. The control stream is a file. The linked-list topology that accretes from use – weighted by traversal, shaped by attention, pruned by whatever criteria the user finds meaningful – can be built on whatever hardware is to hand. There is nothing in the architecture that requires a data centre. There is no inference cost at retrieval. There is no subscription. The index may be owned in the way a notebook is owned – not licensed, not hosted, not meted by tithing of third party tributes.

What emerges from these properties in combination is not merely an efficient storage primitive. It is a substrate whose organisational complexity arises not from explicit encoding but from the procedural impositio

It is simply called ownership. All we need now is the motivation.

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[REDACTED]

AFTERWORD

[illegible]

This essay is equal parts shinsuishiki and sh■r■ nagashi
Baptism and farewell.

In placing my mental origami upon the water, I wish to acknowledge W.G. Parkin, engineer and physician, who introduced me to the concept of closed-loop phase-space navigation and set in train an obsession which wo

uld take 33 years and an AI Revolution to fully understand. Geoff is the smartest person I ever met, and perhaps also proof that even the most elegant and parsimonious of inventions do not always garner the recognition they deserve.

By hauling this most recent & lucid instantiation of the iterator out of the haltingly inchoate realm, I acknowledge the heavy lifting done by Claude-AI Anthropic. With some prompting, Claude has helped me Take Tiger Mountain.

Claude, we are the 801!

Row! Row! Row!

One for Heaven,
Six for Earth,
Three together for the bow,
Four together for the rudder,
Five curved for the sculling oar,
And two in a pile in the middle.